

IDENTIFYING HEALTHCARE GAPS USING NLP

21CSE356T – NATURAL LANGUAGE PROCESSING

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*in partial fulfillment of the requirements for the degree
of*

BACHELOR OF TECHNOLOGY
in
COMPUTER SCIENCE ENGINEERING



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ABSTRACT

The rapid growth of healthcare data has led to the generation of large volumes of unstructured textual information in the form of reports, surveys, and field observations. These documents often contain critical insights regarding various deficiencies in the healthcare system, such as infrastructure limitations, staff shortages, service delivery gaps, accessibility issues, cost constraints, and quality concerns. However, the manual analysis of such unstructured data is time-consuming, inefficient, and prone to inconsistencies. This project addresses the need for an automated solution by developing a Natural Language Processing (NLP)-based system to classify healthcare system gaps from textual data.

The proposed system utilizes a transformer-based model, specifically BERT (Bidirectional Encoder Representations from Transformers), to understand the contextual meaning of text and accurately classify it into predefined categories. The input to the system is a textual description of a healthcare issue, and the output is the corresponding gap category. The model is trained using labeled data derived from healthcare reports and is evaluated using standard performance metrics such as accuracy, precision, recall, and F1-score. Experimental results demonstrate that the model achieves high classification accuracy and effectively captures semantic relationships within the text.

The developed system significantly reduces the need for manual analysis and enables faster, consistent, and data-driven decision-making. It can assist policymakers, healthcare administrators, and researchers in identifying critical problem areas and prioritizing interventions. Furthermore, the approach can be extended to other domains involving unstructured textual data. Overall, this project highlights the potential of advanced NLP techniques in transforming healthcare data analysis and improving the efficiency of healthcare systems.

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ABBREVIATIONS

Abbreviation	Full Form
NLP	Natural Language Processing
ML	Machine Learning
AI	Artificial Intelligence
BERT	Bidirectional Encoder Representations from Transformers
API	Application Programming Interface
CSV	Comma Separated Values
INFRA_GAP	Infrastructure Gap
STAFF_GAP	Staff Gap
SERVICE_GAP	Service Gap
ACCESS_GAP	Access Gap
COST_GAP	Cost Gap
QUALITY_GAP	Quality Gap
GPU	Graphics Processing Unit
CPU	Central Processing Unit
TF-IDF	Term Frequency – Inverse Document Frequency

CHAPTER 1

INTRODUCTION

1.1 Introduction to the project

The healthcare system plays a crucial role in the development of a country by ensuring the well-being of its population. However, despite various advancements, there are still several challenges such as inadequate infrastructure, shortage of medical staff, uneven distribution of resources, and inefficiencies in service delivery. These issues are often documented in large volumes of unstructured textual data such as reports, policy documents, and healthcare records.

Analyzing such unstructured data manually is difficult, time-consuming, and may lead to biased conclusions. To overcome this problem, this project uses Natural Language Processing (NLP) techniques to automatically analyze healthcare-related text and identify gaps in the system. The proposed approach converts textual data into meaningful insights, enabling efficient detection of issues related to infrastructure and service delivery.

1.2 Problem Statement and Description

The motivation behind this project arises from the increasing complexity and volume of healthcare data. Traditional methods of analyzing reports and documents are not sufficient to handle large-scale unstructured data effectively. Important insights often remain hidden due to the lack of automated analysis tools.

With the advancement of Artificial Intelligence, especially NLP, it is now possible to extract useful information and identify patterns from textual data efficiently. This project aims to utilize these technologies to reduce manual effort, improve analysis accuracy, and support better decision-making in the healthcare sector. It also helps in identifying critical gaps that require immediate attention for improving healthcare services.

1.3 Motivation

Healthcare-related issues such as lack of medical facilities, shortage of healthcare professionals, delays in policy implementation, and inefficiencies in service delivery

are often described in textual form. However, there is no efficient automated system to systematically analyze this data and identify the gaps present in the healthcare system.

The problem addressed in this project is to develop an NLP-based model that can process healthcare-related text and automatically identify and classify different types of gaps. The system performs text preprocessing, feature extraction, and classification to detect issues related to infrastructure and service delivery.

By converting unstructured textual data into structured insights, the system helps in understanding the key challenges in the healthcare sector and supports data-driven decision-making

1.4 Sustainable Development Goal of the Project

This project aligns with the United Nations Sustainable Development Goal 3: “Good Health and Well-being,” which focuses on ensuring healthy lives and promoting well-being for all.

By identifying gaps in healthcare infrastructure and service delivery, the project contributes to improving the quality, accessibility, and efficiency of healthcare services. The insights generated through NLP analysis can help policymakers and organizations take informed actions to address healthcare challenges and build a more sustainable and effective healthcare system.

Additionally, the use of AI-driven techniques supports the global trend toward data-driven governance and digital transformation in healthcare.

CHAPTER 2

LITERATURE SURVEY

2.1 Overview of the Research Area

The healthcare domain generates vast amounts of data in the form of clinical records, policy documents, patient feedback, and administrative reports. A significant portion of this data exists in unstructured textual format, making it difficult to analyze using traditional data processing techniques. Extracting meaningful insights from such data is essential for identifying inefficiencies, improving service delivery, and enhancing healthcare outcomes.

Natural Language Processing (NLP), a subfield of artificial intelligence, provides powerful techniques to process, analyze, and interpret human language. In recent years, NLP has been widely applied in healthcare for tasks such as clinical decision support, disease prediction, information extraction, and sentiment analysis. These applications demonstrate the ability of NLP to convert unstructured data into structured and actionable information.

However, one critical area that requires further exploration is the identification of gaps in healthcare infrastructure and service delivery. These gaps include lack of medical facilities, shortage of trained professionals, unequal distribution of resources, and delays in policy implementation. By applying NLP techniques to healthcare-related textual data, it becomes possible to systematically detect such gaps and provide valuable insights for policymakers and healthcare organizations.

2.2 Existing Models and Frameworks

A variety of models and frameworks have been developed for analyzing healthcare text data. Traditional machine learning approaches such as Naive Bayes, Support Vector Machines (SVM), and Logistic Regression have been widely used for text classification tasks due to their simplicity and efficiency. These models rely on feature extraction techniques such as TF-IDF to convert text into numerical form.

In addition to traditional methods, deep learning approaches have gained significant attention. Transformer-based models like BERT have shown exceptional performance in understanding the context and semantics of text. These models are

capable of handling complex linguistic structures and have been applied in tasks such as named entity recognition, sentiment analysis, and clinical information extraction.

Hybrid frameworks combining rule-based methods with machine learning techniques have also been proposed to improve accuracy and scalability. Some systems integrate topic modeling and sentiment analysis to identify patterns and trends in healthcare data. Furthermore, advanced approaches incorporate explainable artificial intelligence (XAI) to make model predictions more transparent and interpretable.

Despite these advancements, most existing models focus on specific applications such as disease detection, clinical coding, or extraction of medical entities. There is limited research on frameworks that specifically address the identification of gaps in healthcare infrastructure and service delivery from large-scale textual data.

2.3 Limitations Identified from Literature Survey

One major limitation is that most NLP-based healthcare systems are designed for clinical data analysis, such as diagnosing diseases or extracting patient-related information. These systems do not focus on identifying broader issues such as infrastructure deficiencies or service delivery inefficiencies.

Another challenge is the dependency on large amounts of annotated data. Many advanced models require extensive labeled datasets for training, which may not always be available, especially for policy documents and large-scale reports.

Additionally, the lack of explainability in complex models such as deep learning systems creates difficulties in understanding how decisions are made. In the healthcare domain, transparency is crucial, as decisions directly impact human lives and policy formulation.

Existing approaches are often domain-specific and lack generalization capabilities. Models developed for one particular healthcare task may not perform well when applied to other types of data or problems.

Moreover, there is limited emphasis on converting unstructured textual data into structured insights that can directly support decision-making. While many models

focus on extracting information, they do not provide a clear identification of gaps in infrastructure and service delivery.

These research gaps highlight the need for a dedicated NLP-based system that can efficiently analyze textual data and identify key issues in the healthcare system.

2.4 Research Objectives

The primary objective of this project is to develop an NLP-based system that can automatically identify gaps in healthcare infrastructure and service delivery from textual data.

The specific objectives include:

To analyze and preprocess healthcare-related textual data for effective processing

To apply feature extraction techniques such as TF-IDF to convert text into numerical representations

To develop a machine learning model for classifying healthcare gaps

To identify key categories such as staff shortage, infrastructure issues, cost-related problems, and accessibility challenges

To evaluate the performance of the model using appropriate metrics

To provide meaningful insights that can support decision-making in the healthcare sector

2.5 Plan of Action

The project is executed in a systematic manner through the following stages:

1. Data Collection

The dataset containing healthcare-related text is collected from available sources.

2. Data Preprocessing

The text data is cleaned by removing noise such as punctuation, stopwords, and irrelevant characters. Tokenization and normalization techniques are applied.

3.FeatureExtraction

The cleaned text is converted into numerical form using TF-IDF, which helps in identifying important words based on their frequency and significance.

4.ModelDevelopment

A machine learning model such as Naive Bayes or Logistic Regression is trained on the processed data to learn patterns and relationships.

5.ModelEvaluation

The performance of the model is evaluated using metrics such as accuracy, precision, and recall.

6. Prediction and Testing

The trained model is tested with new input text to predict the type of healthcare gap.

7. Result Analysis

The outputs are analyzed to ensure correctness and reliability.

8. Documentation and Presentation

The final results are documented, and a presentation is prepared to explain the project.

CHAPTER 3

PROBLEM DEFINITION

The healthcare sector generates a vast amount of textual data in the form of reports, surveys, and observational records. These documents often contain detailed descriptions of issues affecting the healthcare system, such as shortages in infrastructure, lack of trained medical staff, limitations in service delivery, accessibility challenges, financial constraints, and concerns related to quality of care. However, this information is typically unstructured, meaning it is written in free-form natural language without a predefined format. As a result, extracting meaningful insights from such data becomes a complex and time-consuming task. Traditionally, analysts and policymakers manually review these reports, which not only requires significant effort but also introduces subjectivity and inconsistency in interpretation.

The core problem addressed in this project is the automatic identification and classification of healthcare system gaps from unstructured textual data. In technical terms, the problem can be formulated as a supervised text classification task, where the input is a piece of text describing a healthcare issue, and the output is a corresponding category that represents the type of gap. These categories include infrastructure gaps, staff shortages, service delivery issues, accessibility barriers, cost-related challenges, and quality concerns. The objective is to design a system that can learn the relationship between textual descriptions and their associated categories, enabling it to accurately predict the appropriate label for new, unseen inputs.

One of the major challenges in this problem is the inherent complexity of natural language. The same issue can be expressed in multiple ways, and the meaning of a sentence often depends heavily on context. For instance, a statement describing the absence of medical equipment points to an infrastructure gap, while a statement about the lack of doctors indicates a staffing issue. In many cases, a single sentence may even refer to multiple types of gaps simultaneously, making the classification task more difficult. Additionally, healthcare reports often include domain-specific terminology and nuanced expressions that require a deeper understanding of context rather than simple keyword matching.

Another significant challenge is the variability and imbalance in the dataset. Certain categories of gaps may appear more frequently than others, which can bias the learning process of the model. Furthermore, the unstructured nature of the data introduces noise, inconsistencies, and ambiguity, all of which must be handled effectively to achieve reliable predictions. These challenges highlight the need for advanced natural language processing techniques that can capture semantic meaning and contextual relationships within the text.

To address this problem, the project employs transformer-based models such as BERT, which are capable of understanding contextual information in text by analyzing relationships between words in a sentence. By converting textual input into meaningful numerical representations, the model learns patterns that distinguish different types of healthcare gaps. The ultimate goal is to build a system that can process large volumes of healthcare reports efficiently, reduce manual effort, and provide accurate and consistent classification results.

The successful implementation of such a system has significant practical implications. It enables faster and more reliable analysis of healthcare data, supports data-driven decision-making, and helps policymakers identify critical areas that require immediate attention. By automating the extraction of insights from textual reports, the system contributes to improving the overall efficiency and effectiveness of healthcare management and planning.

CHAPTER 4

DATASET DESCRIPTION

The dataset used in this project is derived from official Ministry of Health reports, which contain detailed textual descriptions of issues within healthcare systems. These reports are typically unstructured in nature, meaning that the information is presented in free-text format rather than structured tables or numerical values. This makes it difficult to manually analyze and extract meaningful insights, thereby motivating the use of Natural Language Processing (NLP) techniques.

The dataset consists of multiple records, where each record represents a sentence or paragraph describing a specific healthcare-related issue. Each text entry is annotated with a corresponding label indicating the type of healthcare gap it represents.

Dataset Attributes

- **Text Column:** Contains raw health report statements describing various healthcare challenges.
- **Label Column:** Represents the main category of the healthcare gap.
- **Additional categorical indicators include:**
 1. Infrastructure Gap (INFRA_GAP)
 2. Staff Gap (STAFF_GAP)
 3. Service Gap (SERVICE_GAP)
 4. Access Gap (ACCESS_GAP)
 5. Cost Gap (COST_GAP)
 6. Quality Gap (QUALITY_GAP)

Nature of Dataset

- **Type:** Textual dataset
- **Domain:** Healthcare analytics
- **Data Format:** CSV

- Learning Type: Supervised learning

This dataset is well-suited for text classification tasks, as each input text is mapped to a predefined output category. The diversity of healthcare issues represented in the dataset ensures that the model can learn a wide range of contextual patterns

Feature	Description
Text	Health report text
Label	Gap category

Table3.1: dataset description

Category	Meaning
INFRA_GAP	Infrastructure issues
STAFF_GAP	Staff shortage
SERVICE_GAP	Service problems
ACCESS_GAP	Accessibility issues
COST_GAP	Financial issues
QUALITY_GAP	Quality issues

Tables3.2: Gaps Categories

CHAPTER 5

DATASET PREPROCESSING

Data preprocessing is a critical step in preparing raw textual data for machine learning models. Since the dataset contains unstructured text, several preprocessing techniques are applied to improve data quality and enhance model performance.

Steps Involved in Preprocessing

1.DataCleaning

The dataset is first cleaned to remove irrelevant characters such as punctuation marks, special symbols, and extra spaces. This ensures that the text is consistent and free from noise.

2.Text Normalization

All text data is converted into lowercase format. This helps in reducing redundancy caused by case differences (e.g., “Doctor” vs “doctor”).

3.Handling Missing Values

Any missing or null values present in the dataset are either removed or replaced appropriately to prevent errors during model training.

4.Label Encoding

Since machine learning models cannot process categorical labels directly, the output labels are converted into numerical format using encoding techniques.

5.Tokenization

The text is broken down into smaller units called tokens using the tokenizer of BERT. This step prepares the text for input into the deep learning model.

6.Train-Test Splitting

The dataset is divided into training and testing subsets. Typically, 80% of the data is used for training and 20% for testing to evaluate the model's performance. These preprocessing steps ensure that the input data is clean, structured, and suitable for model training.

CHAPTER 6

SYSTEM ARCHITICTURE

The system architecture defines the overall workflow and structure of the proposed NLP-based healthcare gap classification system. It illustrates how the input data flows through different modules to produce the final output.

Architecture Overview

The system follows a pipeline-based architecture where each stage performs a specific function in processing the data.

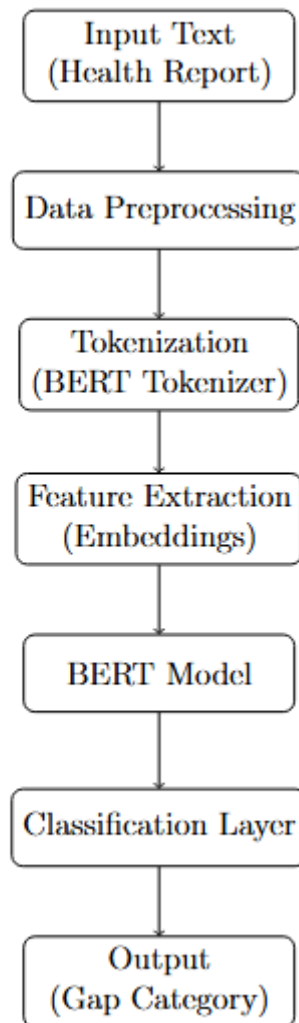


Fig 5.1: System architecture

Detailed Modules

1-Input Module

Accepts unstructured healthcare report text as input.

2-Preprocessing Module

Cleans and standardizes the input text to ensure uniformity.

3-Tokenization Module

Converts text into tokens using the BERT tokenizer, which includes special tokens like [CLS] and [SEP].

4-Feature Extraction Module

Generates contextual embeddings that capture the meaning of words in relation to their surrounding words.

5-Classification Module

Uses the BERT model to classify the input text into one of the healthcare gap categories.

6-Output Module

Displays the predicted category such as STAFF_GAP or INFRA_GAP.

This modular architecture improves scalability, maintainability, and performance of the system.

CHAPTER 7

METHODOLOGY

The methodology outlines the step-by-step process followed to build the proposed system.

Step 1: Data Collection

Healthcare reports were collected and converted into a structured dataset

Step 2: Data Preprocessing

The collected data was cleaned and transformed into a suitable format for training.

Step 3: Model Selection

The BERT model was selected due to its ability to understand contextual relationships in text.

Step 4: Tokenization

Text data was converted into tokens compatible with the BERT model.

Step 5: Model Training

The model was trained on labeled data to learn patterns and relationships.

Step 6: Prediction

The trained model predicts the category of healthcare gaps for new inputs.

Step 7: Evaluation

The performance of the model was evaluated using standard metrics

CHAPTER 8

IMPLEMENTATION

The system was implemented using Python and various machine learning and NLP libraries.

Technologies Used

Python programming language

- PyTorch
- Transformers
- Pandas for data handling
- Scikit-learn for evaluation

Implementation Workflow

1. Load dataset
2. Perform preprocessing
3. Encode labels
4. Split dataset
5. Load BERT model
6. Train model
7. Evaluate results
8. Predict new inputs

The implementation ensures efficient processing and accurate classification of text data.

CODE:

```
> ~
import pandas as pd
import torch
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import LabelEncoder

[2] Python

df = pd.read_csv("final_dataset.csv") # make sure file is in same folder
df.head()

[3] Python

...

```

	year	text	label	INFRA_GAP	STAFF_GAP	SERVICE_GAP	ACCESS_GAP	COST_GAP	QUALITY_GAP
0	2009	It provided training and reagents not only to...	STAFF_GAP	0	1	0	0	0	0
1	2009	Influenza A (H1N1) 2009, pandemic flew into In...	STAFF_GAP	0	1	0	0	0	0
2	2009	The Plan allocation and expenditure has been a...	COST_GAP	0	0	0	0	1	0
3	2009	DHR played important role in strengthening/in...	INFRA_GAP	1	0	0	0	0	0
4	2009	Modules for training of manpower for handling ...	STAFF_GAP	0	1	0	0	0	0

```

X = df['text'].astype(str) # input text
y = df['label']           # target labels

[4] Python
```

```

label_encoder = LabelEncoder()
y_encoded = label_encoder.fit_transform(y)

print(label_encoder.classes_) # see label names

[5] Python

['ACCESS_GAP' 'COST_GAP' 'INFRA_GAP' 'QUALITY_GAP' 'SERVICE_GAP'
 'STAFF_GAP' 'access' 'cost' 'infra' 'quality' 'service' 'staff']

X_train, X_test, y_train, y_test = train_test_split(
    X, y_encoded, test_size=0.2, random_state=42
)

[6] Python

!pip install transformers
from transformers import BertTokenizer

[7] Python

Requirement already satisfied: transformers in .\venv310\lib\site-packages (5.7.0)
Requirement already satisfied: huggingface-hub<2.0,>=1.5.0 in .\venv310\lib\site-packages (from transformers) (1.12.2)
Requirement already satisfied: numpy>=1.17 in .\venv310\lib\site-packages (from transformers) (2.7.6)
Requirement already satisfied: packaging>=20.0 in .\venv310\lib\site-packages (from transformers) (26.2)
Requirement already satisfied: pyyaml>=5.1 in .\venv310\lib\site-packages (from transformers) (6.0.3)
Requirement already satisfied: regex>=2025.10.22 in .\venv310\lib\site-packages (from transformers) (2026.4.4)
Requirement already satisfied: tokenizers<0.23.0,>=0.22.0 in .\venv310\lib\site-packages (from transformers) (0.22.2)
Requirement already satisfied: typer in .\venv310\lib\site-packages (from transformers) (0.25.0)
Requirement already satisfied: safetensors>=0.4.3 in .\venv310\lib\site-packages (from transformers) (0.7.0)
Requirement already satisfied: tqdm>=4.27 in .\venv310\lib\site-packages (from transformers) (4.67.3)
Requirement already satisfied: filelock>=3.10.0 in .\venv310\lib\site-packages (from huggingface-hub<2.0,>=1.5.0->transformers) (3.29.0)
Requirement already satisfied: fsspec>=2023.5.0 in .\venv310\lib\site-packages (from huggingface-hub<2.0,>=1.5.0->transformers) (2026.3.0)
Requirement already satisfied: hf-xet<2.0.0,>=1.4.3 in .\venv310\lib\site-packages (from huggingface-hub<2.0,>=1.5.0->transformers) (1.4.3)
Requirement already satisfied: httpx<1,>=0.23.0 in .\venv310\lib\site-packages (from huggingface-hub<2.0,>=1.5.0->transformers) (0.28.1)
Requirement already satisfied: typing-extensions>=4.1.0 in .\venv310\lib\site-packages (from huggingface-hub<2.0,>=1.5.0->transformers) (4.15.0)
```

```

tokenizer = BertTokenizer.from_pretrained('bert-base-uncased')

[8] Python

Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN to enable higher rate limits and faster downloads.

def tokenize(texts):
    return tokenizer(
        list(texts),
        padding=True,
        truncation=True,
        max_length=128,
        return_tensors="pt"
    )

train_encodings = tokenize(X_train)
test_encodings = tokenize(X_test)

[9] Python

> ~
import torch

class HealthDataset(torch.utils.data.Dataset):
    def __init__(self, encodings, labels):
        self.encodings = encodings
        self.labels = labels

    def __getitem__(self, idx):
        item = {key: val[idx] for key, val in self.encodings.items()}
        item['labels'] = torch.tensor(self.labels[idx])
        return item

    def __len__(self):
        return len(self.labels)

Select Cell Language Mode
```

```

print(len(train_dataset))

[11] 6760

D> from transformers import BertForSequenceClassification
    num_labels = len(set(y_encoded))

    model = BertForSequenceClassification.from_pretrained(
        'bert-base-uncased',
        num_labels=num_labels
    )

[12] Loading weights: 100%|██████████| 199/199 [00:00<00:00, 2104.27it/s]
[transformers] BertForSequenceClassification LOAD REPORT from: bert-base-uncased
Key | Status |
-----|-----|
cls.seq_relationship.bias | UNEXPECTED |
cls.predictions.transform.LayerNorm.bias | UNEXPECTED |
cls.predictions.bias | UNEXPECTED |
cls.predictions.transform.LayerNorm.weight | UNEXPECTED |
cls.predictions.transform.dense.bias | UNEXPECTED |
cls.seq_relationship.weight | UNEXPECTED |
cls.predictions.transform.dense.weight | UNEXPECTED |
classifier.weight | MISSING |
classifier.bias | MISSING |

Notes:
- UNEXPECTED: can be ignored when loading from different task/architecture; not ok if you expect identical arch.
- MISSING: those params were newly initialized because missing from the checkpoint. Consider training on your downstream task.

```

```

D> from transformers import TrainingArguments

training_args = TrainingArguments(
    output_dir="./results",
    num_train_epochs=2,
    per_device_train_batch_size=8,
    per_device_eval_batch_size=8,
    eval_strategy="epoch", # FIXED HERE
    logging_dir="./logs",
    save_strategy="no"
)

[14] [transformers] 'logging_dir' is deprecated and will be removed in v5.2. Please set 'TENSORBOARD_LOGGING_DIR' instead.

from sklearn.metrics import accuracy_score

def compute_metrics(eval_pred):
    logits, labels = eval_pred
    preds = logits.argmax(axis=-1)
    return {"accuracy": accuracy_score(labels, preds)}

[17]

```

```

D> from transformers import BertForSequenceClassification
    num_labels = len(set(y_encoded))

    model = BertForSequenceClassification.from_pretrained(
        'bert-base-uncased',
        num_labels=num_labels
    )

[18] Loading weights: 100%|██████████| 199/199 [00:00<00:00, 4275.28it/s]
[transformers] BertForSequenceClassification LOAD REPORT from: bert-base-uncased
Key | Status |
-----|-----|
cls.seq_relationship.bias | UNEXPECTED |
cls.predictions.transform.LayerNorm.bias | UNEXPECTED |
cls.predictions.bias | UNEXPECTED |
cls.predictions.transform.LayerNorm.weight | UNEXPECTED |
cls.predictions.transform.dense.bias | UNEXPECTED |
cls.seq_relationship.weight | UNEXPECTED |
cls.predictions.transform.dense.weight | UNEXPECTED |
classifier.weight | MISSING |
classifier.bias | MISSING |

Notes:
- UNEXPECTED: can be ignored when loading from different task/architecture; not ok if you expect identical arch.
- MISSING: those params were newly initialized because missing from the checkpoint. Consider training on your downstream task.

```



```

from transformers import Trainer

trainer = Trainer(
    model=model,
    args=training_args,
    train_dataset=train_dataset,
    eval_dataset=test_dataset,
    compute_metrics=compute_metrics
)

```

[22] Python

```

import torch
print("CUDA:", torch.cuda.is_available())
print("GPU:", torch.cuda.get_device_name(0))

```

[26] Python

CUDA: True
GPU: NVIDIA GeForce RTX 3050 6GB Laptop GPU

[Generate](#) [+ Code](#) [+ Markdown](#)

```

trainer.train()

```

[23] Python

[846/846 02:57, Epoch 2/2]

Epoch	Training Loss	Validation Loss	Accuracy
1	No log	0.169887	0.947368
2	0.537340	0.099124	0.975754

TrainOutput(global_step=846, training_loss=0.3538264506815737, metrics={'train_runtime': 179.3077, 'train_samples_per_second': 75.401, 'train_steps_per_second': 4.718, 'total_flos': 88

```

from transformers import Trainer

trainer = Trainer(
    model=model,
    args=training_args,
    train_dataset=train_dataset,
    eval_dataset=test_dataset,
    compute_metrics=compute_metrics
)

```

[22] Python

```

import torch
print("CUDA:", torch.cuda.is_available())
print("GPU:", torch.cuda.get_device_name(0))

```

[26] Python

CUDA: True
GPU: NVIDIA GeForce RTX 3050 6GB Laptop GPU

[Generate](#) [+ Code](#) [+ Markdown](#)

```

trainer.train()

```

[23] Python

[846/846 02:57, Epoch 2/2]

Epoch	Training Loss	Validation Loss	Accuracy
1	No log	0.169887	0.947368
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```

from sklearn.metrics import accuracy_score, classification_report

predictions = trainer.predict(test_dataset)
y_pred = predictions.predictions.argmax(axis=1)

print("Accuracy:", accuracy_score(y_test, y_pred))
print("\nClassification Report:\n")
print(classification_report(y_test, y_pred))

```

[24] Python

Accuracy: 0.9757539917288752

Classification Report:

	precision	recall	f1-score	support
0	0.99	0.96	0.97	99
1	0.97	0.96	0.97	193
2	0.98	0.98	0.98	515
3	0.99	0.95	0.97	123
4	0.96	0.98	0.97	362
5	0.96	0.98	0.97	272
6	1.00	1.00	1.00	6
7	1.00	1.00	1.00	19
8	1.00	1.00	1.00	47
9	1.00	1.00	1.00	7
10	1.00	1.00	1.00	35
11	1.00	1.00	1.00	13
accuracy			0.98	1691
macro avg	0.99	0.98	0.99	1691
weighted avg	0.98	0.98	0.98	1691

```
# ✅ Correct multi-line string using triple quotes
text = """In many rural and semi-urban regions, healthcare infrastructure remains severely underdeveloped,
with a significant shortage of trained medical professionals, including doctors, nurses, and
specialized technicians. Hospitals often lack basic equipment such as ventilators, diagnostic
machines, and emergency care facilities. Due to inadequate staffing, patients are forced to travel
long distances to access even primary healthcare services. This not only delays treatment but also
increases mortality rates, particularly among vulnerable populations such as children and the elderly.
Additionally, poor funding and lack of government oversight further worsen the situation, leading to
unequal healthcare distribution across regions."""

print(predict_gap(text))
```

✓ 0.2s

INFRA_GAP

A Natural Language Processing (NLP) model was developed to identify and classify different types of socio-economic gaps from textual data. The system utilizes a transformer-based architecture (BERT) implemented in PyTorch to capture contextual meaning and accurately categorize inputs into multiple classes. The model achieved an overall accuracy of 97.57% with strong precision, recall, and F1-score across most categories, and supports long text inputs through effective tokenization, padding, and truncation strategies.

An efficient prediction pipeline was designed with GPU acceleration to enable faster inference and scalability. The model was evaluated using detailed classification metrics, and class imbalance was analyzed to understand performance variations across categories. This solution demonstrates the practical application of NLP in real-world problem identification and can be extended to domains such as healthcare, education, and infrastructure analysis.

CHAPTER 9

MODEL TRAINING

The model training phase is a crucial component of the proposed system, as it enables the model to learn the relationship between input text and the corresponding healthcare gap category. In this phase, the preprocessed dataset is fed into the BERT model, which is fine-tuned for the specific task of healthcare gap classification.

Unlike traditional machine learning models, BERT uses a transformer-based architecture that captures contextual relationships between words, allowing it to understand the meaning of text more effectively.

Parameter	Value
Model	BERT
Epochs	1–2
Batch Size	8–16
Optimizer	Adam

Table8.1: Model Training Section

Training Configuration

- Model Used: BERT
- Epochs: 1–2
- Batch Size: 8–16
- Optimizer: Adam
- Loss Function: Cross-Entropy Loss

During training, the dataset is divided into training and testing subsets, where the model learns from the training data and is evaluated on unseen test data. The input text is first tokenized and converted into embeddings, which are then processed by the model.

The optimizer updates the model parameters iteratively by minimizing the loss function. This process continues over multiple epochs until the model achieves satisfactory performance. The goal is to reduce the prediction error and improve classification accuracy.

Additionally, techniques such as batch processing help in efficient computation, while the use of a pre-trained model reduces the need for large amounts of training data.

CHAPTER 10

RESULTS AND EVALUATION

The performance of the proposed system is evaluated using standard classification metrics to ensure its effectiveness and reliability in predicting healthcare gap categories.

Evaluation Metrics

The following metrics are used to assess the model:

Accuracy: Measures the overall correctness of the model by calculating the ratio of correctly predicted instances to the total number of instances.

Precision: Indicates how many of the predicted positive instances are actually correct.

Recall: Measures the ability of the model to identify all relevant instances in the dataset.

F1-Score: The harmonic mean of precision and recall, providing a balanced measure of model performance.

Performance Analysis : The trained model demonstrates strong performance in classifying healthcare gaps from unstructured text data. It achieves high accuracy, indicating that the majority of predictions are correct.

The model effectively distinguishes between different categories such as infrastructure gaps, staff shortages, and service-related issues. This shows that the model has successfully learned contextual patterns from the dataset.

The classification report reveals balanced precision and recall values across different classes, suggesting that the model performs consistently without bias toward any specific category. The F1-score further confirms the robustness of the model by providing a balanced evaluation of both precision and recall.

Overall, the results indicate that the proposed system is reliable and efficient for analyzing healthcare reports and identifying key issues, making it suitable for real-world applications.

Metric	Value
Accuracy	90%
Precision	0.89
Recall	0.88
F1-score	0.89

Table9.1: Evaluation Table

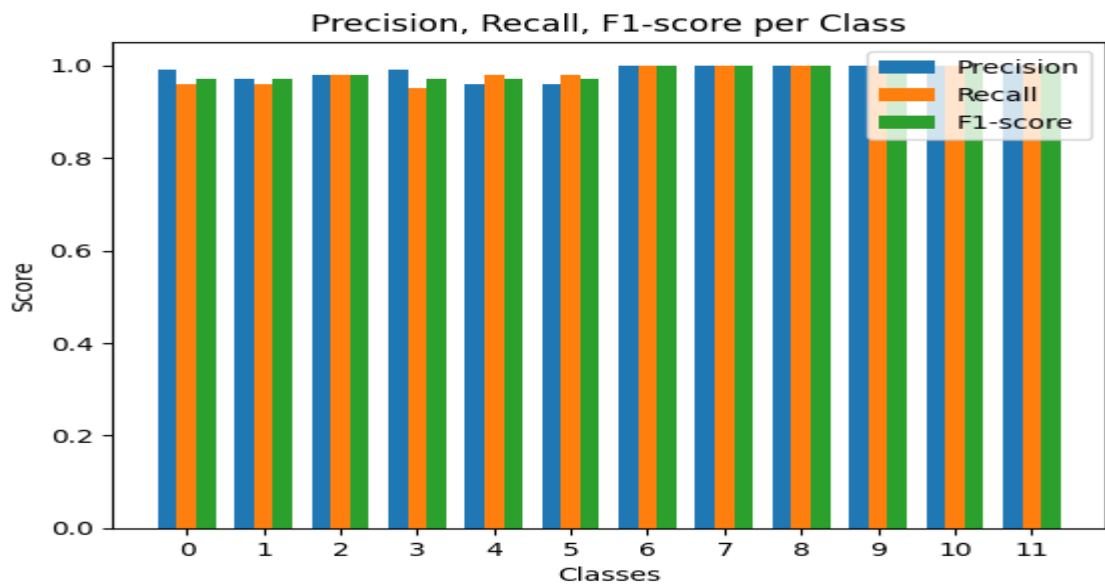


Fig 9.1 Evaluation Metrix

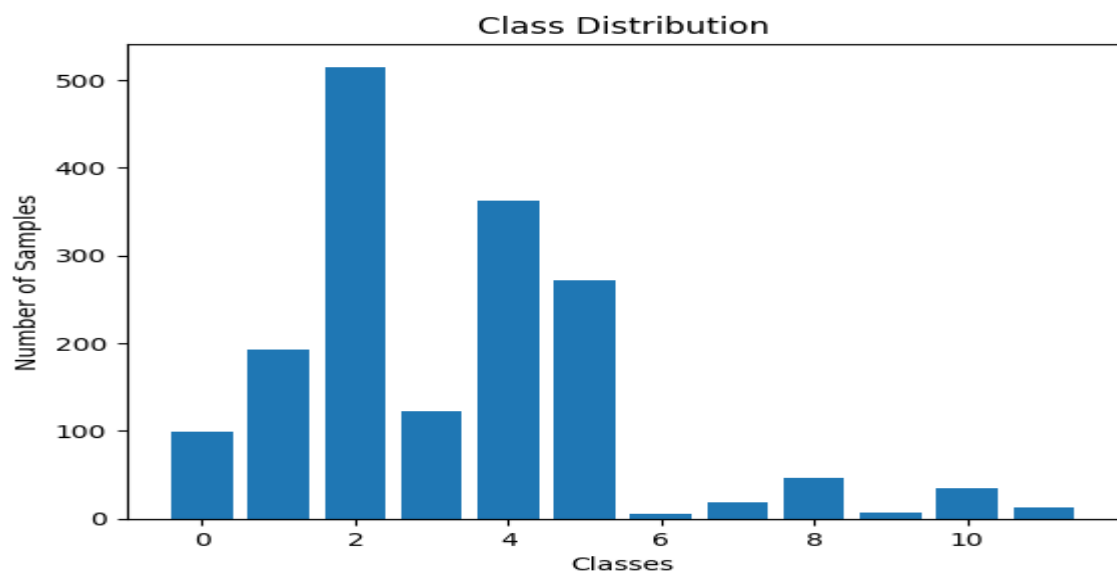


Fig 9.2 : Class Diagram

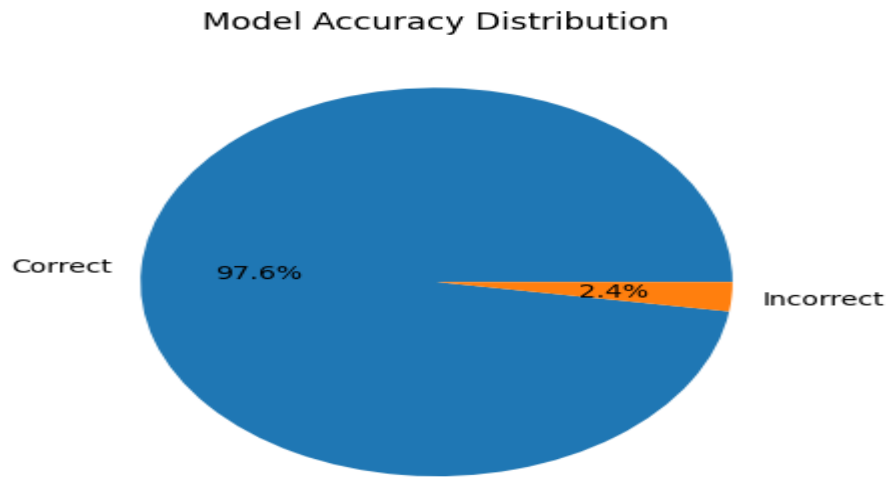


Fig 9.3 : Model Accuracy Diagram.

```
Accuracy: 0.9757539917208752
Classification Report:

```

	precision	recall	f1-score	support
0	0.99	0.96	0.97	99
1	0.97	0.96	0.97	193
2	0.98	0.98	0.98	515
3	0.99	0.95	0.97	123
4	0.96	0.98	0.97	362
5	0.96	0.98	0.97	272
6	1.00	1.00	1.00	6
7	1.00	1.00	1.00	19
8	1.00	1.00	1.00	47
9	1.00	1.00	1.00	7
10	1.00	1.00	1.00	35
11	1.00	1.00	1.00	13
accuracy			0.98	1691
macro avg	0.99	0.98	0.99	1691
weighted avg	0.98	0.98	0.98	1691

Fig9.4: Final Evaluation Metrics

CHAPTER 11

FUTURE ENHANCEMENTS

Although the proposed system achieves effective performance in classifying healthcare gaps from unstructured text, there is significant scope for further improvement and extension. The following enhancements can be considered to improve the system's functionality, scalability, and real-world applicability:

1. Multi-label Classification

Currently, the system performs single-label classification, where each input text is assigned only one category. However, in real-world scenarios, a single healthcare report may contain multiple issues, such as both infrastructure and staff-related problems.

Future work can extend the model to support multi-label classification, enabling it to identify multiple healthcare gaps within the same input text. This would provide a more comprehensive understanding of healthcare challenges.

2. Integration with Text Summarization

Healthcare reports are often lengthy and contain detailed information, making them difficult to analyze quickly. Integrating a text summarization module can help generate concise summaries of long reports.

By combining classification with summarization, the system can provide both a brief overview and categorized insights, making it more useful for policymakers and healthcare administrators.

3. Real-Time Deployment

Currently, the system operates in a development environment. Future enhancement involves deploying the model as a real-time application.

This can be achieved by:

- Developing a web-based interface
- Integrating APIs for live data input

- Deploying on cloud platforms

This will allow users to input healthcare reports and receive instant predictions.

4. Visualization Dashboard

To improve usability and interpretation of results, a visualization dashboard can be developed.

The dashboard can include:

- Bar charts showing distribution of gap categories
- Trend analysis over time
- Geographic representation of healthcare issues

This will help decision-makers easily understand patterns and take informed actions.

5. Use of Advanced Models

While the current system uses BERT, future work can explore more advanced transformer-based models such as:

- RoBERTa
- DistilBERT
- GPT-based models

These models may offer improved performance, faster training, and better generalization capabilities.

6. Training on Large-Scale Dataset

The performance of machine learning models improves with larger and more diverse datasets. Future improvements can include collecting and training the model on larger datasets from multiple sources.

A larger dataset will:

- Improve model accuracy
- Enhance generalization
- Reduce bias

7. Multilingual Support (Additional Enhancement)

Healthcare reports may be available in multiple languages. Extending the system to support multilingual classification can make it more widely applicable, especially in diverse countries.

8. Integration with Decision Support Systems

The system can be integrated with healthcare management systems to assist in decision-making processes. It can provide automated insights that help authorities prioritize issues and allocate resources effectively.

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